4.2 The Early Human Embryo

4.2.1 Introduction: Embryos and Persons

It is very easy to ask the question, 'when does human life begin?' For some people the answer is easy – when the sperm fertilises the egg. For others it is more difficult. Throughout history passionate debate has raged on questions such as follows: at what stage does the duty to protect human life begin, what is entailed by or involved in our duty to protect human life, when is it justifiable to interrupt the fulfilment of human development and so on? The teaching authority of the Roman Catholic (RC) Church and the views of many conservative Protestant Christians (and many 'pro-lifers' in general) currently hold that life begins at fertilisation and that a human life is distinguished from that of animals because humans are 'spiritual beings' (a concept that is also discussed in Chapter 13), a quality that distinguishes us from the rest of the animal kingdom. Interestingly, in the late 20th and early 21st century, spirituality has been embraced as a concept by many people who do not have a religious faith. Used in this non-religious way, it is taken as meaning 'the deepest values and meanings by which people live'[1] and/or the 'essence' of a person's being and the 'inner path' to discovering that essence. But when does this spiritual element actually arise? Or to put it in terms more widely understood, at what stage does the embryo or foetus become a person? Does personhood start at any of the following stages?

- When the sperm and egg first make contact?
- As the sperm gets through the outer layers of the egg?
- When the sperm pronucleus lies alongside that of the egg?
- When the genetic components of the sperm and egg finally unite (i.e. at *syngamy*)?
- During the first cell divisions of embryonic life?[2]
- At the blastocyst stage?
- At implantation?
- At the 'primitive streak' stage (when the beginnings of the nervous system are first laid down)?
- When it ceases to be an embryo and becomes a foetus?[3]
- When the mother first feels the foetus move inside her?
- When the foetus is capable of independent life?

The question of the status of embryos was forced into the public domain in 1978 when Louise Brown, the world's first test-tube baby was born (see Chapter 3). Her mother's egg was fertilised with a sperm from her father in the laboratory where it was allowed to grow for a few days. It was then implanted back into her mother's uterus and the pregnancy proceeded to term when Louise was born. However, this embryo was not the result of a unique feat of embryology. The technique required several eggs to be fertilised in the laboratory and hence the production of several embryos. What was the status of

1 Wikipedia definition.
2 See Figure 4.1.
3 The distinction between embryo and foetus is actually artificial because development is a continuum. According to the US National Institutes of Health, the embryo is called a foetus from week 10 onwards, but some doctors make the terminological change earlier than that (week 8) and others later (week 12).

these 'spare' embryos? Were they human beings? What should be done with them? What protection should the law give them? At the time there were several views in answer to these questions and the debate continues today. The main views are as follows:

- Human life begins at fertilisation.
- We cannot be sure when human life begins but experimentation and other work on the human embryo may reasonably be conducted up to 14 days after fertilisation (as in the terms of the Human Fertilisation and Embryology (HFE) Act, 1990).
- We cannot be sure when human life begins but foetuses up to about 22 weeks are not viable and even babies born prematurely between 22 and 26 weeks are kept alive with great difficulty. This may affect people's views on abortion.
- Human life begins when the baby is born (note that in law, a foetus does not enjoy 'human rights').

We will discuss the first two of these points.

4.2.2 Status of the Embryo: Human Life Begins at Fertilisation

This view is held by many of the so-called pro-lifers. There are two main planks to their argument. Firstly, fertilisation is a specific occurrence (actually it is a series of events that takes several hours) and it is definite – as definite as birth or death. Secondly, the fertilised human egg contains the full genetic complement of the new human being, that is, everything that is genetically necessary for its future growth and development not only as a baby but also into an adult. The pro-life view is summarised in the Encyclical Letter *Evangelium Vitae*, of Pope John Paul II, issued in 1995[4] (but note that this view is not confined to Roman Catholics), '*procured abortion is the deliberate and direct killing by whatever means it is carried out of a human being in the initial phase of his or her existence, extending from conception to birth*'.

Such a view means that:

- No research can be ethically carried out on the early human embryo, including into the causes of genetic disease. To do so would be to use a human being instrumentally, as a means to an end.
- The creation of 'spare' embryos is wrong; there is no such thing as a spare human person (see also Chapter 3).
- Abortion is always wrong and pregnancy must be allowed to continue to birth.
- Infertile couples can never have a child of their own (this issue was discussed in Chapter 3).

Question

For you, is it ethical or unethical to fertilise human eggs outside the uterus for the treatment of infertility? (See also Chapter 3.)

4 Encyclical Letter (25 March 1995), from Pope John Paul II: *Evangelium Vitae*. http://w2.vatican.va/content/john-paul-ii/en/encyclicals/documents/hf_jp-ii_enc_25031995_evangelium-vitae.html (accessed 19 September 2017).

4.2.3 Status of the Embryo: The 14-Day Approach

The egg and the sperm each contain half of their generator's DNA. When the sperm and the egg meet at fertilisation, each fertilised egg (zygote) then contains a full complement of human DNA. For the first few days, the zygote moves within the fallopian tube *en route* to the uterus (Figure 4.1). The genetic identity of a new individual is established at *syngamy*, about 24–30 hours after the initial encounter of the fertilising sperm cell with the egg membrane, and it is at this stage that *cleavage*, or embryonic cell division, begins

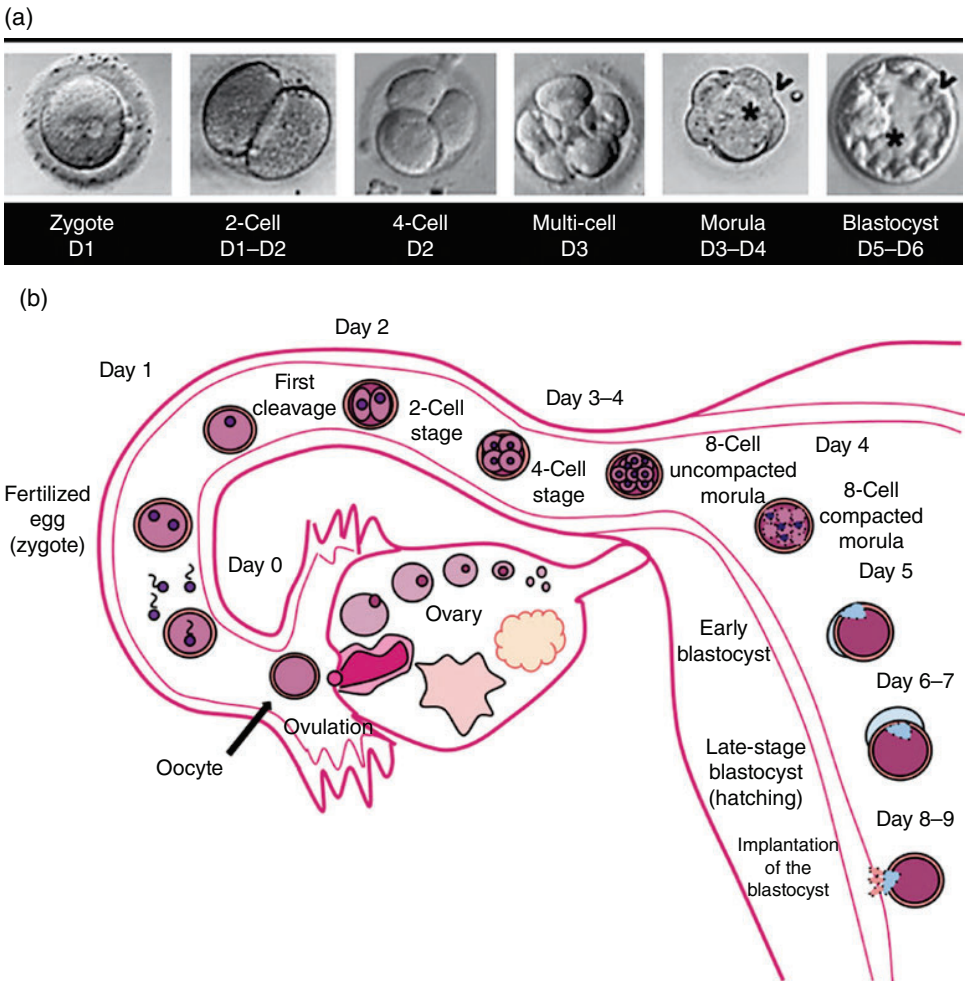

Figure 4.1 (a) Photographs of the early developmental stages of the human embryo. The zygote is the one-cell embryo formed by fertilisation. The two pro-nuclei, one from the egg and one from the sperm, are clearly visible. After merger of the pro-nuclei (syngamy), cell divisions occur; the embryo then becomes compacted (morula stage); in the final stage before implantation, it has hollowed out to form the blastocyst. Key: D1, D2, etc. indicate days of development. *Source:* Photographs are reproduced by permission of Dr Barry Behr, Stanford University School of Medicine. (b) Diagram of human development from fertilisation to implantation. *Source:* Reproduced from Wikipedia under the terms of the Creative Commons licence. https://creativecommons.org/licenses/by-sa/3.0/legalcode.

(see Figure 4.1). The cells divide from two to four to eight and so on, eventually forming a ball of cells called the blastocyst.

The embryo undoubtedly has its own unique version of the human genome. But is the moral status of the embryo based on just on genetic uniqueness? What then is the status of certain aberrantly formed embryos? For example, it has been known for some years that human eggs can undergo cleavage in the absence of fertilisation. This condition is known as parthenogenesis, and in an experimental situation, development beyond the primitive streak stage has been observed from unfertilised mouse eggs. Would we ascribe personhood to a parthenogenetically dividing embryo? Further, if the possession of a unique human genome is a major criterion, what can be said about 'identical' twin embryos, let alone cloned embryos?

The blastocyst (Figure 4.1) starts to form at about five to six days and when it has about 30–60 cells (which is at eight to nine days), it may attach itself to the wall of the uterus. At about this time, the embryo can divide into two and become identical twins: this is of course the natural production of clones. If attachment to the wall of the uterus does occur, then a pregnancy is established. However, in humans, 70–80% of fertilised eggs/early embryos simply do not survive to establish a pregnancy. There is a huge natural wastage.

Question

Can these failed embryos be described as human beings?

Attachment to the uterine wall leads to hormonal changes in the mother; she misses her monthly period and may begin to 'feel pregnant'. The blastocyst's inner cell mass will form the 'embryo proper' (which later becomes the foetus) and the outer cells will become the placenta. At about 14 days the cells become more organised and a strip of cells called the primitive streak forms from which nervous system will develop. However, some embryos detach and if this happens the woman is said to have suffered an early miscarriage (but the woman may not know even know that she was ever pregnant). By 10–11 weeks, organs are formed and early limbs are also present.

In summary, between fertilisation and 14 days after fertilisation, the survival of the fertilised egg is in doubt and many do not survive. Survival depends on the blastocyst embedding/implanting itself in the wall of the uterus but even at that stage, some are lost. Furthermore, cells in the early embryo may develop either into a baby (or two babies if twinning occurs) or the placenta, which is discarded at birth. Finally, there is no evidence of a nervous system before 14 days and a functioning nervous system is an essential part of our being. Thus, it is argued that it is acceptable to use embryos for research up to the 14-day stage and that view is currently embedded in UK and US law (but see Section 4.3).

4.3 Embryo Research

It was obvious after the birth of Louise Brown (see previous chapter) that test-tube babies or assisted reproduction would become a standard option in the treatment of infertility. The ability to fertilise eggs in the laboratory and grow embryos also opened

up many other possibilities: research into the causes of infertility, miscarriage and genetic disease, identifying genetic disease and selecting only healthy embryos for implantation into the uterus. As mentioned in the previous chapter, the UK government set up a committee, chaired by the distinguished philosopher Mary Warnock, to examine the science and the ethics of these possibilities and make recommendations. Its report, published in 1984, known as the Warnock Report, was the basis for the legislation that currently governs this area of biomedical research and practice. The Report adopted the '14-day' approach to the human embryo discussed in the previous section. It is a classic piece of principled pragmatism. It recognised that the early human embryo was indeed human material but, according to the Report, could not yet be regarded as a human person because firstly many embryos in nature never reach implantation and secondly the rudiments of the central nervous system are not present. To give it the same protection as a human person would be to prevent any research into diseases that cause much suffering. However, scientists were not to be free to do whatever they liked with human embryos. Their activities are governed by the Human Fertilisation and Embryology Authority (HFEA) set up by Act of Parliament in 1990. A minority report published alongside the main Warnock Report[5] took the more 'traditional' view that human life begins at fertilisation and indeed that view received significant support in Parliament during debates on the bill as it went through the stages of becoming an act.

The HFE Act has five main provisions:

1) All activities related to human reproductive technologies are supervised by the HFEA and may only be conducted under a licence granted by the Authority.
2) Embryo donation and donor insemination are allowed under licence. The Act also sets out who are the lawful parents of children conceived by artificial reproduction and the confidentiality arrangements about the genetic parents, although, as was noted in the previous chapter, the law on anonymity has since been changed.
3) Embryos and gametes may be frozen. They may be used in the future with the consent of the donors. They are destroyed after ten years.
4) The legislative arrangements for surrogacy are set out.
5) Research on human embryos may be carried out up to 14 days after fertilisation, under licence for the following purposes:
   i) Promoting advances in understanding infertility
   ii) Gaining knowledge about congenital disease
   iii) The development of contraceptive methods
   iv) Detecting gene and chromosomal abnormalities

As new developments arise, the HFEA rules on whether or not they may be pursued, granting the appropriate licence (some examples are discussed in Chapters 5 and 6).

5 Discussed by Department of Health and Social Security (1984) *Report of the Committee of Inquiry into Human Fertilisation and Embryology (The 'Warnock Report')*. HMSO, London; Bulger RE et al., eds (1995) *Society's Choices: Social and Ethical Decision Making in Biomedicine*. National Academies Press, Washington, DC.

Question

The Warnock Committee's recommendations depended heavily on our scientific knowledge about fertilisation and the development of the human embryo during the first 14 days. How far should scientific knowledge contribute to making ethical decisions?

The whole debate about whether or not embryo research is morally acceptable really centres on the definition of an embryo, and this is again related to the question of when life begins. In this respect, the arguments *against* research using embryos are similar to those against abortion. For those people who believe that a unique human person comes into being at the fertilisation, any procedure that interferes with the normal development of the embryo is unacceptable because they see that embryo as having the same rights and interests as any other child or adult.

Experimental procedures on human embryos are only permitted under the law for the first 14 days after the mixing of the gametes – in other words before the appearance of the primitive streak. Even then, under the law, certain types of research on human embryos were actually prohibited. These include:

- Inserting a human embryo into an animal.
- Nucleus substitution – This procedure comprises removal of the nucleus of an embryonic cell and replacing with the nucleus from a cell of another embryo or person. This restriction has now been lifted for two specific purposes, namely, for research on 'therapeutic cloning' (Chapter 5) and for IVF involving mitochondrial donation (Chapter 3).
- Altering the genetic structure of any cell while it forms part of an embryo. This restriction has now been lifted specifically to allow mitochondrial donation in IVF.
- Cloning of human embryos for reproductive purposes.

Other than in these circumstances, it is only after this period of 14 days from fertilisation that protected human life legally begins. An embryo that has been experimented upon cannot be maintained *in vitro*, frozen or replaced into a woman after that time, and it must be destroyed.

However, there has recently been a call to extend the limit to 21 or even days. In May 2016, two research groups reported that they had managed to get human blastocysts to implant into an artificial uterine lining and that the implanted blastocysts began to self-organise and differentiate.[6] This was a major achievement and suggested that embryos could be kept alive and developing normally for longer than 14 days (although both research groups terminated their experiments before 14 days). It is suggested that being able to use embryos up to 21 days after fertilisation will enable research on early developmental lesions and on early miscarriage.

6 www.cam.ac.uk/research/news/scientists-develop-human-embryos-beyond-implantation-stage-for-first-time (accessed 19 September 2017); Deglincerti A et al. (2016) Self-organization of the *in vitro* attached human embryo. *Nature* **533**, 251–254.

There is a lot of support for this change,[7] but some have expressed caution,[8] while the 'pro-life' movement is (obviously) against it.

Recent reports suggest that there is real hope of improvement in mobility for patients with spinal injuries, following treatments arising from stem cell research.[9]

Question

Should more embryos be made available for research into this sort of treatment? If so, where they should they come from?

In the early years of this century, the HFEA granted its first licences for the use of human embryos in stem cell research. This was a significant moment for British science, and it has opened the door for research into a wide range of hitherto incurable diseases. This is discussed more fully in Chapter 5. It has long been held in the United Kingdom that the sensitivity of feeling about human embryo research is such that the development of any new technique must be strictly regulated. Research licences are only granted when it can be demonstrated that the researchers will add to the body of knowledge and in doing so gain technical competence. Only then can they be allowed to use a new technique in the treatment of patients. It is with this careful and painstaking approach coupled with ongoing public debate that big issues – and embryonic stem cell research can surely be said to be one of the most significant in recent years[10] – can be approached responsibly. There is no doubt that research will continue to strive to improve on nature: indeed, discussion of the elimination of 'undesirable' genes is already one of the consequences of the Human Genome Project, but this should always be tempered with questions such as 'what actually is meant by undesirable?'

4.4 Screening and Diagnosis

In several countries, including the United Kingdom, a pregnant woman is offered a series of screening tests (in the United Kingdom, paid for by the NHS) at an antenatal clinic. Ultrasound scans and blood tests screen for abnormalities, for example, spina bifida and Down's syndrome. Ethically, although these tests do not harm the foetus, they may lead to moral dilemmas for the parents. For example, if the test for Down's syndrome is positive, a decision on whether or not to proceed with the pregnancy will have to be considered. The screening tests are not compulsory and some women decide not to have them.

7 E.g. www.theguardian.com/commentisfree/2016/may/06/extend-14-day-limit-embryo-research (accessed 19 September 2017).
8 Thus, the Nuffield Council on Bioethics has commissioned the preparation of a research paper on medical and ethical aspects of the proposal.
9 See Chapter 5.
10 But see Chapter 5.